

Summary

Researcher and engineer with 5+ years building open-source data infrastructure and ML systems on heterogeneous real-world data. PhD from TU Eindhoven on interpretable ML and multi-source time-series data, with peer-reviewed publications in *Journal of Building Engineering* and *Energy Efficiency*; dual M.Sc. (Hons) Mathematics and B.E. (Hons) Civil Engineering. Designer and maintainer of ComfortOS, an open-source platform unifying sensor, weather, and survey streams behind a server-driven schema with reproducible experiments built in. Strong graph theory background, hands-on with Python, modern ML, and LLM integration, with a track record of shipping open-source tooling and peer-reviewed research in multi-institutional consortia.

Experience

Smart Building Engineer / Data & Controls at [Spectral](#), Amsterdam Sept 2025 – Present

- Build production data pipelines ingesting BMS, IoT, weather, and sub-metering streams behind a common semantic layer, so downstream ML and analytics teams can reuse heterogeneous data without re-ingesting at the source
- Engineer FastAPI services, Python clients, and cloud-deployed components end-to-end, working across data, models, and product teams

PhD Researcher – Open Data Infrastructure & ML for Multi-Source Sensor Data at the [Brains4Buildings](#) consortium, TU Eindhoven Sept 2021 – August 2025

- Designed and deployed [ComfortOS](#), an open-source multi-tenant platform (FastAPI, PostgreSQL, Docker) with a unified connector gateway and server-driven schema, making heterogeneous research data findable, accessible, interoperable, and reusable across consortium partners (TU/e, TU Delft, TNO, industry)
- Built interpretable ML models (PyTorch, SHAP, partial dependence, surrogates) on multi-source time-series datasets, including occupant survey data collected via a self-developed iOS app, with reproducibility baked into the platform
- Published peer-reviewed work in *Journal of Building Engineering* and *Energy Efficiency*; supervised Master's students through data projects; tutored and guest-lectured on smart-building methodology

Full Stack Developer, Q3 Technologies, India Aug 2020 – July 2021

- Built production web applications and REST APIs (Python/Django, React) on AWS for enterprise clients; designed data models, authentication, and CI/CD pipelines with test automation

Founding Engineer, [GPLAN.in](#) Jan 2018 – May 2020

- Co-developed graph-theoretic and genetic-algorithm methods at a research startup spun out of BITS Pilani; shipped a production [web application](#) and co-authored peer-reviewed publications in *Automation in Construction* and *Computer Graphics Forum*

Education

2021 - 2025  **PhD** - Department of the Built Environment, TU Eindhoven, Netherlands.

2015 - 2020  **M.Sc. (Honours) Mathematics** BITS Pilani, India.

 **B.E. (Honours) Civil Engineering** BITS Pilani, India.

Tech Stack

- Python (PyTorch, scikit-learn, pandas, NumPy) • SQL • JavaScript • Git • CI/CD • Linux
- FastAPI • PostgreSQL • Docker • REST APIs • Multi-Source Data Integration • Schema Design
- Graph Theory • Combinatorial Optimisation • Machine & Deep Learning • Interpretable ML
- LLM Integration • Knowledge Representation • FAIR Principles • Open-Source Research Tooling

Applications & Tools Developed

ComfortOS	Open-source multi-tenant data platform with FastAPI backend and unified connector gateway. Integrates heterogeneous real-time streams into a common queryable model for cross-site analytics.
P3Venti	Decision-support web dashboard built for long-term care centres to balance ventilation, infection risk, and energy use, used by domain stakeholders without engineering backgrounds.
Self-Reporting App	iOS app collecting structured occupant survey data via scheduled notifications, feeding a curated time-series dataset for ML research.
GPLAN Game	Cross-platform iOS/Android app translating graph-based optimisation algorithms into an interactive educational game.

Highlighted Projects

ComfortOS | Open-Source FAIR-Aligned Data Platform

- Designed and deployed [ComfortOS](#), a multi-tenant FastAPI/PostgreSQL/Docker platform with a unified connector gateway and server-driven schema, ingesting heterogeneous real-time streams (BMS, IoT, weather, occupant surveys) into a common queryable model. Direct precursor to the kind of FAIR research infrastructure needed for cross-disciplinary data reuse.
- Open-source with reproducible experiment pipelines and per-site configuration; lets researchers without engineering backgrounds find, access, and rebuild experiments on shared data.

Graph-Based & Combinatorial Optimisation | Published Research, Production App

- Co-developed graph-theoretic and variable-chromosome genetic algorithm (VCGA) methods for floorplan generation and Steiner-tree pipeline routing (NP-hard) for rainwater harvesting tank optimisation on CNN-classified satellite imagery. Strong foundation in graph data structures, the same primitives that underpin RDF, SPARQL traversal, and knowledge graphs. Published in *Automation in Construction* and *Computer Graphics Forum*.

Interpretable ML for Multi-Source Time-Series Data

- Developed deep learning and tree-based forecasting models (72% accuracy) on time-series sensor and survey data collected via a self-developed [iOS app](#); used SHAP, partial dependence, and surrogate models to interpret model behaviour and ground decisions in transparent evidence. Published in *Energy Efficiency* and *Journal of Building Engineering*.

Teaching, Workshops & Supervision

2022–2024	Master Project Supervision & Guest Lecturer, TU/e – Supervised Master’s students through research design, data analysis, and results; tutor and guest lecturer for Smart Building Methodology and Technology.
2020	CAADRIA 2020 Workshop – Facilitated hands-on workshop on graph-theoretic generative design and optimisation at the international conference.
2019	MATLAB Workshop, BITS Pilani – Designed and led sessions on optimisation, neural networks, and structural dynamics for Master’s and PhD students.

Select Publications

1. **N. Upasani**, O. Guerra-Santin, and M. Mohammadi, “[Developing building-specific, occupant-centric thermal comfort models: A methodological approach](#),” *Journal of Building Engineering*, vol. 95, 2024.
2. **N. Upasani**, O. Guerra-Santin, M. Mohammadi, M. Seraj, and F. Joosstens, “[Understanding thermal comfort using self-reporting and interpretable machine learning](#),” *Energy Efficiency*, 2025.
3. **N. Upasani**, K. Shekhawat, and G. Sachdeva, “[Automated Generation of Dimensioned Rectangular Floorplans](#),” *Automation in Construction*, vol. 113, 2020.
4. S. Bisht, K. Shekhawat, **N. Upasani**, R. Jain, R. Tiwaskar, and C. Hebbar, “[Transforming an Adjacency Graph into Dimensioned Floorplan Layouts](#),” *Computer Graphics Forum*, vol. 41(6), 2022.

The full list of publications is available [here](#).